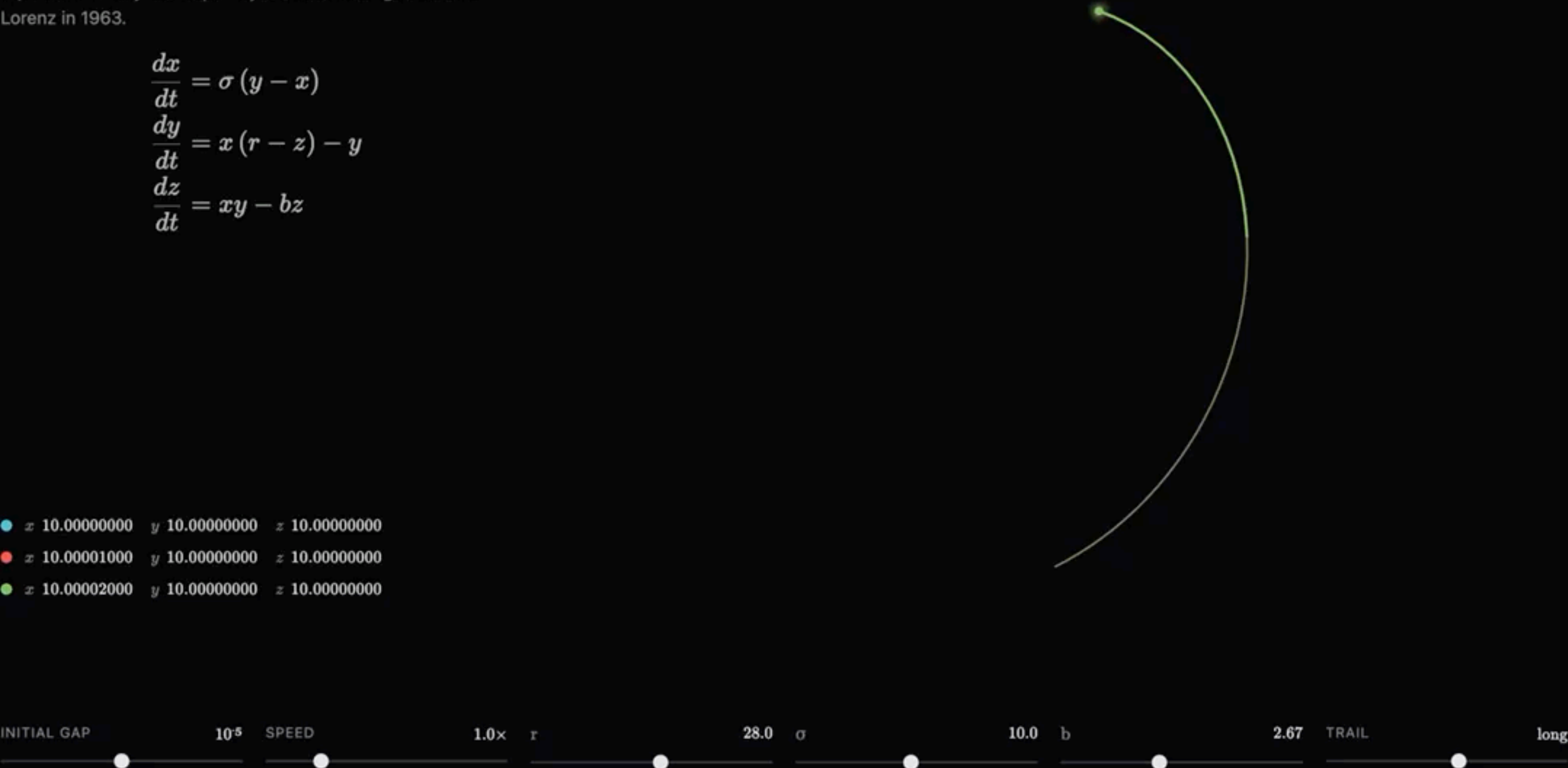


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- x 10.00000000 y 10.00000000 z 10.00000000
- x 10.00001000 y 10.00000000 z 10.00000000
- x 10.00002000 y 10.00000000 z 10.00000000

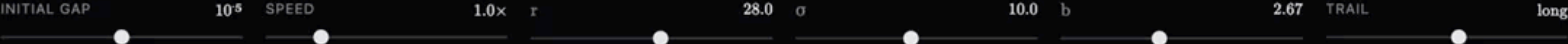


TIME
0.14

SEPARATION
 1.59×10^5

CHAOS THRESHOLD τ_H
24.74

$r = 28.0 > 24.7$ — chaotic

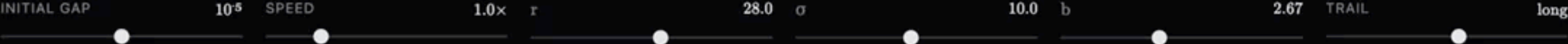


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- x 10.00000000 y 10.00000000 z 10.00000000
- x 10.00001000 y 10.00000000 z 10.00000000
- x 10.00002000 y 10.00000000 z 10.00000000



TIME
0.14

SEPARATION
1.59×10⁵

CHAOS THRESHOLD τ_H
24.74

r = 28.0 > 24.7 — chaotic

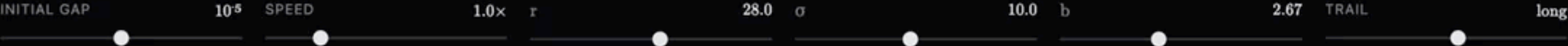


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- x 10.000000000 y 10.000000000 z 10.000000000
- x 10.00001000 y 10.000000000 z 10.000000000
- x 10.00002000 y 10.000000000 z 10.000000000



Trapping Lemma for the Lorenz System

Let

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(r - z) - y, \\ \dot{z} &= xy - bz.\end{aligned}$$

Lemma. *There exists a bounded ellipsoid E that is a trapping region for the Lorenz flow. Equivalently, the time-one map T of the Lorenz flow satisfies*

$$T(E) \subset \text{int}(E).$$